MARKOV CHAINS & BROWNIAN MOTION — Master Sheet

Ch. 1 & Ch. 6 (Durrett)

1. Markov Property & Transition Matrix

Markov property: for state space S,

$$\mathbb{P}(X_{n+1}=j \mid X_n=i, X_{n-1}, \dots) = \mathbb{P}(X_{n+1}=j \mid X_n=i) = p(i,j).$$

$p(i,j) \geq 0$, $\sum_j p(i,j) = 1$ (rows of P sum to 1).

Distribution at time n: row vector $\nu_n = \nu_0 P^n$.
n-step: $p^n(i,j) = (P^n)_{ij}$.
Chapman–Kolmogorov: $p^{m+n}(i,j) = \sum_k p^m(i,k)p^n(k,j)$, i.e. $P^{m+n} = P^m P^n$.
Verifying a chain: write X_n in terms of underlying r.v.s; if conditioning on X_{n-1} alone is *not enough* to fix the distribution of X_n (older info needed), it is *not* Markov. Cue: same X_{n-1} producing different next-step distributions $\Rightarrow$ NOT a chain.
Building P from a story: list states $\to$ enumerate cases for each transition $\to$ multiply probabilities of independent draws. Verify each row sums to 1.

2. State Classification

Reach: $i \to j$ iff $\exists\, n \geq 0 :\ p^n(i,j) > 0$.
Communicate: $i \leftrightarrow j$ iff $i \to j$ and $j \to i$ (equiv. rel. $\Rightarrow$ classes).
Closed set C: no exits; $\sum_{j \in C} p(i,j) = 1\ \forall i \in C$.
Irreducible: one class.
Absorbing state: $p(i,i) = 1$.
Transient: $\mathbb{P}(\text{return} \mid X_0 = i) < 1 \Rightarrow \mathbb{E}[\text{visits}] < \infty$.
Recurrent: $\mathbb{P}(\text{return} \mid X_0 = i) = 1$.
Period: $d(i) = \gcd\{n \geq 1 : p^n(i,i) > 0\}$. Aperiodic $\Leftrightarrow d = 1$. Test: any self-loop $\Rightarrow d = 1$, else find two cycles of coprime length.
Decomposition (finite): $S = T \cup R_1 \cup \cdots \cup R_k$, T transient, each R_ℓ irreducible closed recurrent.
Finite-chain facts:

- Every closed set contains a recurrent state.
- Every recurrent class is closed.
- Transient states never form a closed set.
- Irreducible finite chain $\Rightarrow$ all states recurrent.

Procedure: draw the directed graph from P; find SCCs; an SCC with no outgoing edges is closed/recurrent; others transient.

3. Stationary Distribution π

Definition: $\pi P = \pi$, $\pi_i \geq 0$, $\sum_i \pi_i = 1$.
Linear system: solve $\pi(P - I) = \mathbf{0}$ with $\sum \pi_i = 1$. Drop one row of $\pi(P-I) = 0$ (rank-deficient), replace with normalization. Row-reduce.
Detailed balance (reversibility): $\pi_i\, p(i,j) = \pi_j\, p(j,i)\ \forall i,j \Rightarrow \pi$ stationary. Use whenever the graph is a tree, birth-death, or random walk on a graph. *Sufficient, not necessary.*
Existence/uniqueness:

- Irreducible finite chain: unique π, $\pi_i > 0$.
- Reducible: each recurrent class R_ℓ has its own $\pi^{(\ell)}$ supported on R_ℓ. Set of stationary distributions = convex hull $\{\sum \alpha_\ell \pi^{(\ell)}\}$.

Mean return time: for irreducible positive recurrent chain, $\mathbb{E}_i[T_i] = 1/\pi_i$ where $T_i = \min\{n \geq 1 : X_n = i\}$.
Two-state shortcut: $P = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right) \Rightarrow \pi = \left(\frac{b}{a+b}, \frac{a}{a+b}\right)$.

4. Convergence to Equilibrium

Convergence theorem (irreducible + aperiodic): $p^n(i,j) \to \pi_j$ for all i, j.
Periodic case: $\lim p^n$ DNE; but Cesaro avg $\frac{1}{n}\sum_{k=1}^n p^k(i,j) \to \pi_j$ still.
Bipartite ($d = 2$): $P = \left(\begin{smallmatrix} 0 & A \\ B & 0 \end{smallmatrix}\right) \Rightarrow P^2 = \left(\begin{smallmatrix} AB & 0 \\ 0 & BA \end{smallmatrix}\right)$. Find u with $u(AB) = u$; set $v = uA$ (then $v(BA) = v$). Stationary of P: $\frac{1}{2}(u, v)$. Stationary set of P^2: $\{\alpha(u,0) + (1-\alpha)(0,v) : \alpha \in [0,1]\}$.
Convergence test: (1) one communicating class *and* (2) aperiodic. Verify both.

5. Hitting Probabilities & Expected Times

Let $A \subset S$, $T_A = \min\{n \geq 0 : X_n \in A\}$.
Hit prob $h_i = \mathbb{P}_i(T_A < \infty)$:

- $h_i = 1$ for $i \in A$;
- $h_i = \sum_j p(i,j) h_j$ for $i \notin A$;
- $h_i = 0$ if i in a closed set disjoint from A.

Expected hit time $\tau_i = \mathbb{E}_i[T_A]$: $\tau_i = 0$ for $i \in A$; $\tau_i = 1 + \sum_j p(i,j)\tau_j$ for $i \notin A$.
General $\lim p^n(i,j)$:

1. Identify recurrent classes $R_1, \dots, R_k$.
2. Compute $h_i^{(\ell)} = \mathbb{P}_i(\text{absorbed in } R_\ell)$ for each transient i.
3. Compute stationary $\pi^{(\ell)}$ on each R_ℓ.
4. $\lim p^n(i,j) = h_i^{(\ell)} \cdot \pi_j^{(\ell)}$ when $j \in R_\ell$ (R_ℓ aperiodic).

6. Two-State Convergence

For $P = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$, $p_n = \mathbb{P}(X_n = 1)$. Markov $\Rightarrow$

$$p_{n+1} = (1-a)p_n + b(1-p_n) = (1-a-b)p_n + b.$$

Iterate / geom. series ($r = 1-a-b$):

$$\boxed{p_n = \tfrac{b}{a+b} + (1-a-b)^n\left(p_0 - \tfrac{b}{a+b}\right).}$$

Converges *exponentially* to $b/(a+b)$ when $a+b < 2$. Rate: $|1-a-b|$. Trick: subtract steady-state from both sides; the fixed point cancels.

7. Special Chains

Gambler's Ruin
$\{0, \dots, N\}$, 0 & N absorbing. $p(i,i+1) = p$, $p(i,i-1) = q = 1-p$. Ruin: $\mathbb{P}_i(T_0 < T_N) = \frac{(q/p)^i - (q/p)^N}{1-(q/p)^N}$ $(p \neq q)$, $\frac{N-i}{N}$ $(p = q = \frac{1}{2})$.

Birth–Death Chain
$p(i,i+1) = p_i$, $p(i,i-1) = q_i$, $p(i,i) = r_i$. Detailed balance: $\pi_{i+1} = \pi_i\, p_i/q_{i+1}$, so $\pi_i = \pi_0 \prod_{k=0}^{i-1} p_k/q_{k+1}$.

Random Walk on Graph
$p(i,j) = 1/\deg(i)$ if $i \sim j$. Stationary: $\pi_i = \deg(i)/(2|E|)$. Detailed balance auto.

Ehrenfest / Urn Exchange
$2m$ balls total, b black; state i = #black in urn 1. $p(i,i+1) = \frac{(m-i)(b-i)}{m^2}$, $p(i,i-1) = \frac{i\,(m-b+i)}{m^2}$, $p(i,i) =$ rest. Stationary:

$$\pi(i) = \binom{b}{i}\binom{2m-b}{m-i}\Big/\binom{2m}{m} \quad \text{(hypergeometric)}.$$

Intuition: equilibrium urn 1 = uniform random size-m subset of all $2m$ balls.

Coupon Collector / Dice
Roll n-sided die; X_n = #distinct values seen. $p(i,i+1) = (n-i)/n$, $p(i,i) = i/n$. $\mathbb{E}[T_{i\to i+1}] = n/(n-i) \Rightarrow \mathbb{E}[T_n] = nH_n$. $n = 6 \Rightarrow 6H_6 \approx 14.7$.

8. Stationary-Distribution Recipes

A. Full system. Stack $\begin{bmatrix} P^\top - I \\ \mathbf{1}^\top \end{bmatrix}\pi = \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}$, solve.

B. Detailed balance. Try $\pi_i p(i,j) = \pi_j p(j,i)$; if consistent, telescope along a spanning tree.

C. Block / bipartite. $P = \left(\begin{smallmatrix} 0 & A \\ B & 0 \end{smallmatrix}\right) \Rightarrow$ solve $u(AB) = u$, $v = uA$.

D. Multi-class. Solve on each recurrent class. Convex hull is full set.

9. Common Pitfalls

- "Closed" $\neq$ "absorbing" (set vs. single state).
- Detailed balance is sufficient, not necessary.
- Don't forget normalization $\sum \pi_i = 1$.
- For periodic chain, use Cesaro avg or block-decompose P^d.
- Transient j does *not* appear in $\lim p^n(\cdot, j)$.
- Multi-class: weight $\pi_j^{(\ell)}$ by hit prob $h_i^{(\ell)}$.
- "Same start, different futures" $\Rightarrow$ not Markov.

10. Independent Normals (Ch. 6.10)

If $X \sim N(\mu_1, \sigma_1^2) \perp Y \sim N(\mu_2, \sigma_2^2)$: $aX + bY \sim N(a\mu_1 + b\mu_2,\ a^2\sigma_1^2 + b^2\sigma_2^2)$.
If $X, Y \sim N(0, \sigma^2)$ iid, $U = \frac{X+Y}{2}$, $V = \frac{X-Y}{2}$ are iid $N(0, \sigma^2/2)$ since $\mathrm{Cov}(U,V) = \frac{1}{4}[\mathrm{Var}\, X - \mathrm{Var}\, Y] = 0$ and *joint normal + uncorrelated $\Rightarrow$ independent.*
Linear-combo identities:

- $\mathbb{E}[aX + bY] = a\mathbb{E}X + b\mathbb{E}Y$ (always linear).
- $\mathrm{Var}(aX + bY) = a^2\,\mathrm{Var}\, X + b^2\,\mathrm{Var}\, Y + 2ab\,\mathrm{Cov}(X,Y)$.
- $\mathrm{Cov}(aX + bY, Z) = a\,\mathrm{Cov}(X,Z) + b\,\mathrm{Cov}(Y,Z)$.
- Uncorrelated $\not\Rightarrow$ independent generally; for joint Gaussians it does.

11. Brownian Motion (Ch. 6.11a)

Definition (standard BM on $[0,\infty)$):
1. $B(0) = 0$;
2. Independent increments: for $0 \leq t_0 < \cdots < t_k$, $B(t_i) - B(t_{i-1})$ are mutually independent;
3. Gaussian increments: $B(t) - B(s) \sim N(0, t-s)$, $t > s$;
4. Sample paths $t \mapsto B(t)$ continuous a.s.

Key formulas: $\mathbb{E}[B(t)] = 0$, $\mathrm{Var}[B(t)] = t$, $\mathrm{Cov}(B(s), B(t)) = \min(s,t)$. The vector $(B(t_1), \dots, B(t_k))$ is jointly Gaussian, $\Sigma_{ij} = \min(t_i, t_j)$.
Symmetries: $-B(t)$, $cB(t/c^2)$ (Brownian scaling), $tB(1/t)$ (time inversion, with $B(0) := 0$) are all standard BMs. Markov property: $B(t+\cdot) - B(t)$ is BM independent of $\{B(s) : s \leq t\}$.
Lévy / dyadic construction on $[0,1]$: let $\{Y_{n,m}\}$ iid $N(0,1)$. Set $B(0) = 0$, $B(1) = Y_{0,1}$. Recursively for $0 \leq m \leq 2^n - 1$:

$$B\left(\tfrac{2m+1}{2^{n+1}}\right) = \tfrac{1}{2}\left[B\left(\tfrac{m+1}{2^n}\right) + B\left(\tfrac{m}{2^n}\right)\right] + 2^{-n/2-1}Y_{n,m}.$$

Why this matters: BM is defined on dyadic rationals with the right increment law; continuity follows from a.s. uniform convergence of the random series (Kolmogorov / Borel–Cantelli).

12. Normal Tail Bound (Ch. 6.9)

For $0 < x \leq y$:

$$(1 - 3y^{-4})e^{-y^2/2} \leq e^{-y^2/2} \leq e^{-x^2/2}e^{-x(y-x)}.$$

Left: $1 - 3y^{-4} \leq 1$ since $y^{-4} \geq 0$.
Right: take logs $\Rightarrow (y-x)^2 \geq 0$.
Integrate over $y \in [x, \infty)$:

$$\boxed{\left(\tfrac{1}{x} - \tfrac{1}{x^3}\right)e^{-x^2/2} \leq \int_x^\infty e^{-y^2/2}dy \leq \tfrac{1}{x}e^{-x^2/2}}$$

Tricks: LHS antiderivative – combine $\frac{d}{dy}\left(-\frac{e^{-y^2/2}}{y}\right) = \frac{e^{-y^2/2}}{y^2} + e^{-y^2/2}$ and $\frac{d}{dy}\left(\frac{e^{-y^2/2}}{y^3}\right)$ to get the $-3y^{-4}$ piece. RHS: $u = y - x \Rightarrow \int_0^\infty e^{-xu}du = 1/x$.
Mills' ratio corollary: $\mathbb{P}(Z > x) \sim \frac{1}{x\sqrt{2\pi}}e^{-x^2/2}$ as $x \to \infty$.

13. Absorbing-Chain Framework

Reorder states: transient T first, absorbing A last. Then

$$P = \begin{pmatrix} Q & R \\ 0 & I \end{pmatrix}, \quad Q\colon T \to T, \quad R\colon T \to A.$$

Fundamental matrix: $N = (I - Q)^{-1} = \sum_{k=0}^\infty Q^k$. Then $N_{ij} = \mathbb{E}_i[\#\text{visits to } j]$ for $i, j \in T$.
Expected time to absorption: $t = N\mathbf{1}$, i.e. $t_i = \sum_{j \in T} N_{ij}$.
Absorption probabilities: $B = NR$, with $B_{ia} = \mathbb{P}_i(\text{absorbed in } a)$ for $i \in T$, $a \in A$.
Useful for: gambler's ruin, mixed-class limits, any "probability you eventually reach j" question with multiple recurrent components.

14. Eigendecomposition & Mixing Rate

If $P = UDU^{-1}$, then $P^n = UD^nU^{-1}$. Eigenvalues $1 = \lambda_1 \geq |\lambda_2| \geq \cdots$. For irreducible aperiodic finite chain, $|\lambda_2| < 1$ and

$$\|p^n(i,\cdot) - \pi\|_{TV} \leq C|\lambda_2|^n,$$

so $|\lambda_2|$ controls the mixing rate. Spectral gap $1 - |\lambda_2|$ governs convergence speed. Sanity: 2×2 chain has eigenvalues 1 and $1 - a - b$ – matches the closed form $p_n - \frac{b}{a+b} \propto (1-a-b)^n$.
For *reversible* chain (π + detailed balance), P is similar to a symmetric matrix $D^{1/2}PD^{-1/2}$ (with $D = \mathrm{diag}(\pi)$) and has all real eigenvalues in $[-1, 1]$.

15. Quick-Reference Table

Prompt	Method
"Is this a chain?"	Check $P(X_n \mid X_{n-1})$ self-suff.
"Transition matrix"	Enumerate cases; rows sum to 1
"ν_n"	$\nu_0 P^n$
"Long-run fraction"	$\pi P = \pi$, normalize
"Hit prob $\to A$"	$h_A = 1$, $h_i = \sum p(i,j) h_j$
"Expected time $\to A$"	$\tau_A = 0$, $\tau_i = 1 + \sum p(i,j)\tau_j$
"Mean return time"	$1/\pi_i$ (irreducible)
"$\lim p^n$ multi-class"	$h_i^{(\ell)} \cdot \pi_j^{(\ell)}$
"Bipartite lim"	Block P^2, π on each block
"Use detailed balance?"	Tree / birth-death / RW on graph
"Convergent?"	Irreducible + aperiodic
"Period?"	gcd of return-times; self-loop $\Rightarrow d = 1$
"Time to absorption"	$t = (I - Q)^{-1}\mathbf{1}$
"Absorption probs"	$B = (I - Q)^{-1} R$
"Mixing rate"	$\lvert\lambda_2\rvert^n$ decay
"BM increment var"	$t - s$
"Sum of normals"	$N(\sum \mu, \sum \sigma^2)$ if indep
"Tail of Z"	Mills: $\sim \phi(x)/x$

16. Probability & Algebra Toolbox

$\mathbb{E}[g(X)] = \sum_x g(x)p(x)$ or $\int g(x)f(x)dx$. Linearity always.
$\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$, $\mathrm{Cov}(X,Y) = \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y$.
Conditional: $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]]$, $\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$.
Geometric series: $\sum_{k=0}^{n-1} r^k = \frac{1-r^n}{1-r}$, $\sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$ $(|r| < 1)$.
Harmonic: $H_n = \sum_{k=1}^{n} 1/k \approx \ln n + \gamma$.
Indicator trick: $\mathbb{E}[\#\text{visits to } j] = \sum_{n=0}^{\infty} p^n(i,j)$. Transient $\Leftrightarrow$ this sum is finite.
Hypergeometric: $\mathbb{P}(X = k) = \binom{K}{k}\binom{N-K}{n-k}/\binom{N}{n}$, mean nK/N.

17. Numpy Snippets

Stationary distribution:
```
n=P.shape[0]
A=np.vstack([P.T-np.eye(n),
np.ones(n)])
b=np.append(np.zeros(n),1)
pi=np.linalg.lstsq(A,b,
rcond=None)[0]
```
n-step distribution:
```
nu=nu0 @ np.linalg.
matrix_power(P,n)
```
Hit prob into class A:
```
Q=P[T][:,T]; R=P[T][:,A]
N=np.linalg.inv(
np.eye(len(T))-Q)
B=N@R  #absorption probs
t=N@np.ones(len(T))  #times
```
Eigenvalues / mixing:
```
w,v=np.linalg.eig(P.T)
lam2=sorted(np.abs(w))[-2]
```

18. Last-Minute Sanity Checks

- Each row of P sums to 1.
- Stationary π has $\pi_i \geq 0$, $\sum \pi_i = 1$.
- Irreducible chain $\Rightarrow$ $\pi_i > 0$ strictly.
- Multi-class chain $\Rightarrow$ π on recurrent set.
- Limit matrix: transient column $= 0$.
- Detailed balance check: re-do on ≥ 1 extra edge.
- Hit-prob system: $h = 1$ on target, $h = 0$ on competing closed sets.
- BM: any normal-increment problem reduces to "$N(0, \Delta t)$".